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Manufacturing process with atomic level inspection

Abstract

Costs may be avoided and yields improved by applying scanning probe microscopy to substrates in the midst of an integrated circuit fabrication process sequence. Scanning probe microscopy may be used to provide conductance data. Conductance data may relate to device characteristics that are normally not available until the conclusion of device manufacturing. The substrates may be selectively treated to ameliorate a condition revealed by the data. Some substrates may be selectively discarded based on the data to avoid the expense of further processing. A process maintenance operation may be selectively carried out based on the data.

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Background/Summary

REFERENCE TO RELATED APPLICATIONS (1) This application is a Divisional of U.S. application Ser. No. 17/199,879, filed on Mar. 12, 2021, which claims the benefit of U.S. Provisional Application No. 63/142,562, filed on Jan. 28, 2021 and U.S. Provisional Application No. 63/150,234, filed on Feb. 17, 2021. The contents of the above-referenced patent applications are hereby incorporated by reference in their entirety.

BACKGROUND

(1) Fabricating integrated circuits typically includes processing a substrate such as a semiconductor wafer through a large number of fabrication processes to form various features and devices. Substrates are put through hundreds of fabrication processes, which may include, but are not limited to, lithographic processes, plasma etching, wet etching, chemical vapor deposition (CVD), sputter deposition, chemical-mechanical polishing (CMP), ion implantation, annealing, variations thereof, and the like.

(2) There is an ongoing demand for progressively higher device density. As a result, some fabrication processes may be operated close to the limits of their capabilities. Due to the high number of devices that are processed and the pushing of process limits, some fraction of the manufactured devices may be imperfect. To assure quality, manufacturers perform inspections on finished devices and discard those that fail to meet the manufacturer's standards. There is a cost associated with discarding devices and an ongoing need to reduce costs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 illustrates a cross-sectional side view a manufacturing system according to some aspects of the present teachings.

(3) FIG. 2 illustrates a scanning probe microscope that may be used in accordance with the present teachings.

(4) FIG. 3 is a flow chart of a method according to some aspects of the present teachings.

(5) FIG. 4 illustrates a cross section of a memory device at an intermediate stage of manufacturing.

(6) FIGS. 5-7 illustrates cross sections of the memory device of FIG. 4 at subsequent stage of manufacturing.

(7) FIG. 8 illustrates a cross section of an integrated circuit at an intermediate stage of manufacturing.

(8) FIG. 9 illustrates a cross section of another integrated circuit at an intermediate stage of manufacturing.

(9) FIG. 10 is a plot off C-AFM data obtained in accordance with some aspects of the present

teachings.

(10) FIG. 11 is a flow chart of a method according to some aspects of the present teachings.

(11) FIG. 12 is an image of data obtained in accordance with some aspects of the present teachings.

(12) FIGS. 13A-C are black and white images derived from the image of FIG. 12 in accordance with other aspects of the present teachings.

(13) FIG. 14 is a flow chart of a method according to some other aspects of the present teachings.

(14) FIG. 15 is a flow chart of a method according to some other aspects of the present teachings.

DETAILED DESCRIPTION

(15) The present disclosure provides many different embodiments, or examples, for implementing different features of this disclosure. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(16) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(17) Inspections are typically carried out at the conclusion of integrated circuit manufacturing to determine electrical characteristics such as write voltage, read voltage, current, and durability. There is a disadvantage, however, in that these inspections are time consuming and costly. According to the present teachings, costs may be avoided and yields improved by applying scanning probe microscopy to substrates in the midst of an integrated circuit fabrication process sequence. In some embodiments, substrates are selectively treated to ameliorate a condition detected by the microscopy. In some embodiments, substrates are selectively discarded to avoid the expense of further processing. In some embodiments, the scanning probe microscopy provides conductance data. The conductance data may relate to device characteristics that are normally not available until the conclusion of device manufacturing.

(18) Some aspects of the present teachings relate to a method that includes applying a first manufacturing process to a substrate and then scanning a portion of the substrate surface with a scanning probe microscope to obtain data. The data is used to make a diagnostic determination on the basis of which a second manufacturing process is selectively either applied or not applied to the substrate. In some embodiments, the second manufacturing process is an ordinary part of the manufacturing sequence that is only carried out if the diagnostic determination established that the result of prior processing is satisfactory. In some embodiments, the second manufacturing process is a remedial process by which a condition reflected by the diagnostic determination may be ameliorated. In some embodiments, a third manufacturing process is applied while the data is being analyzed. Continuing processing in parallel with the data analysis may improve throughput.

(19) In some embodiments, the scanning probe microscope comprises an atomic force microscope that collects topological (height) data. In some embodiments, the scanning probe microscope comprises a conductance atomic force microscope that collects conductance data. In some embodiments, the scanning probe microscope provides both topological data and conductance data. Conductance data may directly relate to electrical properties that are normally evaluated only after

fabrication has been completed.

(20) In some embodiments, the portion of the substrate surface that is scanned is only a fraction of the substrate surface. In some embodiments, a plurality of discrete areas of the substrate surface are included in the scan. Scanning a plurality of discrete areas may account for center to periphery variations and other variations that may occur systematically across a substrate surface. In some embodiments, a plurality of scanning probes are used to increase the scanning rate. In some embodiments, different sample areas are selected from one substrate to the next in order increase a probability of detecting a defect that could be systematically localized.

(21) Making a diagnostic determination involves processing the data. In some embodiments, processing the data includes rendering the data as an image. In some embodiments where there is more than one type of data, the data is rendered as a plurality of images. In some embodiments, one image represents topological data and another image represents conductance data. The data may be filter, scaled, consolidated, or otherwise processed as it is rendered into one or more images.

(22) In some embodiments, processing the data comprises identifying device images that correspond to individual instances of a device that has multiple instances across the image. In some embodiments, the device is a memory cell. In some embodiments, the device is a metal gate. In some embodiments, a diagnostic determination is made based on the data within the device images (grey-scale image data). In some embodiments, device shape data is extracted from the grey-scale image data and the diagnostic determination is made based on the shape data. In some embodiments, perimeters of the device images are identified and the diagnostic determination is made based on the shapes of the perimeters.

(23) In some embodiments, the first process includes ion beam etching. In some embodiments, the first process includes a wash process that follows the ion beam etching and prepares the substrate for a subsequent deposition process. The diagnostic determination may establish whether conductive residues are present in amounts that could cause excessive shorting. In some embodiments, an additional etch or clean process is performed selectively based on the diagnostic determination.

(24) In some embodiments, the first process is a spacer etch or the like. In some embodiments, the first process includes a wet clean that follows the spacer etch and prepares the substrate for a subsequent deposition process. In some embodiments, the spacer is a first spacer formed adjacent a memory cell. In some embodiments, the spacer is a second spacer formed adjacent a memory cell. In some embodiments, the memory cell comprises a metal tunneling junction (MTJ). The diagnostic determination may establish whether spacer coverage is too great or too little. In some embodiments, an additional etch process is selectively performed based on the diagnostic determination. In some embodiments, an additional deposition process is selectively performed based on the diagnostic determination.

(25) In some embodiments, the first process is a planarization process. In some embodiments, the first process includes a cleaning process that follows a planarization process. In some embodiments, the planarization process comprises chemical mechanical polishing (CMP). In some embodiments, the diagnostic determination establishes whether an electrode has been adequately exposed by CMP. In some embodiments, the electrode is a top electrode of a memory cell. In some embodiments, the electrode is of a high-x metal gate (HKMG) transistor. In some embodiments, the electrode is for one or more of a bipolar junction transistor (BJT), an n-channel metal oxide semiconductor (nMOS) transistor, a p-channel metal oxide semiconductor (pMOS) transistor, fin field effect transistors (finFETS), a gate-all-around FET (GAAFET), a gate-surrounding FET, a multi-bridge channel FET (MBCFET), a nanowire FET, a nanoring FET, a nanosheet field-effect transistor (NSFET), or the like. In some embodiments, a remedial CMP process is performed selectively based on the diagnostic determination. In some embodiments, the diagnostic determination establishes whether an excessive amount of conductive residue is present on the substrate. In some embodiments, an additional etch or clean process is performed selectively based

on the diagnostic determination.

(26) Some aspects of the present teachings relate to an integrated circuit device manufacturing system that includes a first processing tool, a second processing tool, a substrate handling system, a scanning probe microscope, and a computer processor. The scanning probe microscope receives and scans substrates that have been processed in the first processing tool. The computer processor receives data from the scanning probe microscope and issues instructions to the substrate handling system on the basis of the data. The substrate handling system may be positioned upstream from the second processing tool and may selectively deliver substrates to the second processing tool according to the instructions from the computer processor. In some embodiments, some substrates that are not immediately delivered to the second processing tool are redirected for remedial processing. In some embodiments, some substrates that are not delivered to the second processing tool are redirected for disposal or recycling.

(27) FIG. 1 illustrates a manufacturing system **100** according to some aspects of the present teachings. The manufacturing system **100** includes a substrate handler **107**, a scanning probe microscope **113**, a computer processor **101**, and one or more processing tools such as a first tool **109**, a second tool **111**, a third tool **115**, and a fourth tool **119**. The substrate handler **107** may receive substrates **105** from a loading system **103** and selectively move them to any of the first tool **109**, the second tool **111**, the third tool **115**, the fourth tool **119**, and the scanning probe microscope **113**. The substrate handler **107** may receive instructions from the computer processor **101**. The computer processor **101** may receive data from the scanning probe microscope **113**.

(28) The processing tools such as the first tool **109**, the second tool **111**, the third tool **115**, and the fourth tool **119** can each be any of the processing tools used in the integrated circuit manufacturing industry. Examples of processing tools used in the integrated circuit manufacturing industry include, without limitation, sputtering tools, vapor deposition tools (including tools for chemical vapor deposition, atomic layer deposition, and plasma enhanced deposition), plasma etching tools, lithography tools, wet chemical processing tools, polishing tools, ion beam etching systems, furnaces, and the like.

(29) As illustrated by FIG. 2, the scanning probe microscope **113** may be a conductive atomic force microscope (C-AFM). The scanning probe microscope **113** includes a cantilever **207** having a tip **209** positioned to scan a surface **215** of a substrate **105** on a stage **211** of a table **213**. The table **213** may comprise an XY or XYZ positioning stage. The stage **211** moves the substrate **105** under the tip **209**, whereby the tip **209** travels across the surface **215**. Alternative, the tip **209** may be configured to move while the stage **211** remains stationary. A power source **206** is operative to maintain a voltage difference between the stage **211** and the substrate **105**. An amp meter **205** measures a current between the tip **209** and surface **215**. The current data may be conveyed to the computer processor **101** or an intermediary processing device.

(30) The scanning probe microscope **113** may further include a system for measuring deflection of the cantilever **207** such as a photodiode **217** configured to detect light **203** reflected off the cantilever **207** by the laser **201** and configured to measure deflection of the cantilever **207**. The photodiode **217** may convey its data to the computer processor **101** or the intermediary device. Accordingly, the scanning probe microscope **113** is also an atomic force microscope (AFM) that measures variations in height across the surface **215**.

(31) Although the scanning probe microscope **113** has been illustrated as a C-AFM, the scanning probe microscope **113** may be any type of scanning probe microscope. Alternatively or in addition to being a C-AFM, the scanning probe microscope **113** may be on or more of an atomic force microscope (AFM), a chemical force microscope (CFM), an electrostatic force microscope (EFM), a Kelvin probe force microscope (KPFM), a magnetic force microscope (MFM), a piezo-response force microscope (PFM), a photothermal microscope, a scanning capacitance microscope (SCM), a scanning gate microscope (SGM), a scanning voltage microscope (SVM), a scanning tunneling microscope (STM), or the like. The scanning probe microscope **113** may have a contacting mode of

operation, a non-contacting mode of operation, or both contacting and non-contacting modes of operation. The scanning probe microscope **113** has been illustrated as applying a voltage to the substrate **105** through the stage **211**. Alternatively, the voltage may be applied directly to the surface **215** of the substrate **105**.

(32) The substrate **105** may include one or more semiconductor, conductor, and/or insulator structures. In some embodiments, the substrate **105** is in the form of a wafer such as a semiconductor wafer. A semiconductor may be an elementary semiconductor such as silicon or germanium with a crystalline, polycrystalline, amorphous, and/or other suitable structure; a compound semiconductor such as silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, indium antimonide, or the like; an alloy semiconductor such as SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, GaInAsP or the like; or a combination thereof. Combinations of semiconductors may take the form of a mixture or gradient such as a substrate in which the ratio of Si and Ge vary across locations. The substrate **105** may include a layered semiconductor. A layered semiconductor may include a semiconductor layer on an insulator such as that used to produce a silicon-on-insulator (“SOI”) substrate, a silicon-on-sapphire substrate, or a silicon-germanium-on-insulator substrate, or a layer of semiconductor on glass such as that used to produce a thin film transistor (“TFT”). The manner of coupling the voltage to the substrate **105** may be selected according to the substrate type.

(33) In some embodiments, the surface **215** of the substrate **105** includes at least one material that is conductive, at least one material that is insulating, and optionally other materials. Devices including one or more of these materials may be formed or partially formed on or over the substrate **105**. In some embodiments, the substrate **105** is undergoing front-end-of-line processing (FEOL) and the devices are in or immediately above the substrate **105**. In some embodiments, the substrate **105** is undergoing back-end-of-line processing (BEOL) and the devices are being formed within a metal interconnect structure above the substrate **105**. In some embodiments, the devices include at least one electrode. In some embodiments, the electrodes are metal. In some embodiments, the devices are memory cells. In some embodiments, the devices are transistors. Other device types that may be at the surface **215** include, without limitation, diodes, photocells, resistors, capacitors, semiconductor fins, and the like. In some embodiments, the conductive material on the surface **215** forms conductive lines or vias.

(34) FIG. 3 provides a flow chart of a method **300** that may be used with the manufacturing system **100**. The method **300** begins with act **301**, carrying out a first process. The first process may be a process carried out on the substrate **105** by the first tool **109**. The method **300** continues with act **303**, scanning probe microscopy to inspect a surface **215** of the substrate **105**. Act **303** provides data such as height data or conductivity data. The substrate handler **107** may transfer the substrate **105** from the first tool **109** to the scanning probe microscope **113** for the scanning.

(35) The method **300** continues with act **307**, performing an analysis of data from the scanning probe microscopy and using the results to make a diagnostic determination. While the evaluation is taking place, the method **300** may simultaneously continue with an optional act **305**, applying an interim process. The interim process may be a process carried out on the substrate **105** by the second tool **111**. The substrate handler **107** may transfer the substrate **105** from the scanning probe microscope **113** to the second tool **111** for this interim processing.

(36) The method **300** continues with act **309**, determining whether to carry out a remedial process based on the diagnostic determination. In some embodiments, the diagnostic determination is expressed in terms of a score and act **309** determined whether that score exceeds a threshold value. In some embodiments, the score relates to a degree of uniformity of devices on the substrate **105**. For example, the score may be based on conductance ranges and correlate with a uniformity in threshold voltages. In some embodiments, the score relates to a degree of conformity of devices on the substrate **105** to a specification. For example, the score may relate to a percentage of devices on the substrate **105** that have a leakage path with conductivity above a predetermined threshold. In

some embodiments, the diagnostic determination provides a particular indication if a threshold condition has been met.

(37) If act **309** determines that remediation is appropriate the method **300** may continue with act **311**, which is carrying out a remedial process. The remedial process may be a process carried out on the substrate **105** by the third tool **115**. The substrate handler **107** may transfer the substrate **105** from the scanning probe microscope **113** or from the second tool **111** for this remedial processing. If act **309** determines that remediation is not appropriate or if the remediation of act **311** is complete, the method **300** may continue with act **313**, which is carrying out the subsequent process. The subsequent process may be a process carried out on the substrate **105** by the fourth tool **119**. The substrate handler **107** may transfer the substrate **105** from the scanning probe microscope **113**, from the second tool **111**, or from the third tool **115** to the fourth tool **119** for the subsequent process to be carried out.

(38) In some embodiments, the first process is ion beam etching. In some embodiments, the first process is a wet clean that follows the ion beam etching. In some embodiments the wet clean is an interim process. FIG. 4 illustrates a cross section **401** of a substrate **105A** that includes memory cells **402** immediately following ion beam etching. The memory cells **402** may include a top electrode **403**, a magnetic tunneling junction (MTJ) **405**, and a bottom electrode **407**. The top electrode **403** and the bottom electrode **407** are metal and are conductive. The bottom electrodes **407** may be coupled by vias **411** to conductive lines or vias **417** in an underlying metal interconnect layer **419**, wherein the conductive lines or vias **417** are separated from one another by a dielectric material **423**, which may be a low- κ dielectric. A barrier layer **415**, such as tantalum nitride or titanium nitride, as well as one or more etch stop layers **421**, such as a silicon nitride layer, may also be present. The metal interconnect layer **419** may be the third, the fourth, the fifth, or some of other metal interconnect layer within a metal interconnect structure of the substrate **105A**. A surface **215A** of the substrate **105A** at this stage of processing includes the memory cells **402** and a dielectric layer **413**. The MTJ **405** may include a lower ferromagnetic layer **405a** and an upper ferromagnetic layer **405c**, which are separated from one another by an insulator layer **405b**. In some embodiments, the insulator layer **405b** is a tunnel barrier and may be sufficiently thin to allow carriers to tunnel between the lower ferromagnetic layer **405a** and the upper ferromagnetic layer **405c**.

(39) In some embodiments, the surface **215A** is scanned using C-AFM. Although the topographical variation appears to be distinctive, it has been found the C-AFM has less noise and provides more reliable diagnostic determinations than AFM for this application. In some embodiments, a diagnostic determination identifies residues or a particle on the surface **215A** and the remedial process includes further cleaning. In some embodiments, a diagnostic determination identifies excessive conductivity on a sidewall **420** of a memory cell **402** or a like location, the excessive conductivity being of a type that may be caused by redeposition of conductive material, and the remedial process includes etching adapted to remove redeposited conductive material. In some embodiments, a diagnostic determination identifies a sidewall angle that is too shallow and the remedial process includes additional etching to steepen the sidewall angle. In some embodiments, the diagnostic determination indicates the desirability of scrapping or recycling the substrate **105A**.

(40) In some embodiments, the first process is a spacer etch. In some embodiments, the first process is a wet clean that follows the spacer etch. In some embodiments the wet clean is an interim process. In some embodiments, the spacer etch is a first spacer etch. FIG. 5 illustrates a cross section **501** of the substrate **105A** immediately following a first spacer etch. At this stage of processing, a surface **215B** of the substrate **105A** includes an exposed portion of the top electrodes **403**, spacers **503**, which are insulators, and the dielectric layer **413**. In some embodiments, the spacer etch is a second spacer etch that forms a second spacer over a first spacer. FIG. 6 illustrates a cross section **601** of the substrate **105A** immediately following a second spacer etch. The second spacers **603** are formed over the spacers **503** and may be separated from the spacers **503** by a liner

layer **605**. At this stage of processing, a surface **215C** of the substrate **105A** includes an exposed portion of the top electrodes **403**, second spacers **603**, which are dielectric, and the dielectric layer **413**.

(41) The surface **215B** or the surface **215C** may be scanned using AFM, C-AFM, or both AFM and C-AFM. C-AFM may be particularly effective for determining an extent to which the top electrodes **403** have been exposed. The diagnostic determination may indicate whether the spacer etch has underexposed or overexposed the top electrodes **403**. In some embodiments, a remedial process includes additional etching. In some embodiments, a remedial process includes an additional deposition of spacer material, which may be followed by an additional etch process.

(42) In some embodiments, the first process is a polishing process such as chemical mechanical polishing (CMP). In some embodiments, the first process is a wet clean that follows the CMP. In some embodiments the wet clean is an interim process. In some embodiments, the CMP process exposes top electrodes of memory cells. FIG. 7 illustrates a cross-sectional view **701** of the substrate **105A** immediately following CMP. At this stage of processing, a surface **215D** of the substrate **105A** includes an exposed portion of the top electrodes **403**, spacers **503**, which are insulators, the dielectric layer **413**, a dielectric etch stop or CMP stop layer **703**, interlevel dielectric **705**, and conductive plugs **707**. The interlevel dielectric **705** may be a low-K dielectric or an extremely low-K dielectric. The surface **215D** may be scanned using AFM, C-AFM, or both AFM and C-AFM. C-AFM may be particularly effective for determining an extent to which the top electrodes **403** have been exposed. The diagnostic determination may determine if CMP has underexposed or overexposed the top electrodes **403**. In some embodiments, a remedial process includes additional CMP. In some embodiments, a remedial process includes an additional deposition of dielectric, which may be followed by an additional CMP process.

(43) In some embodiments, the CMP process exposes top electrodes of transistor gates. FIG. 8 illustrates a cross section **801** of a substrate **105B** that includes a first metal gate layer **803** and a second metal gate layer **805** that together provide a metal gate electrode over a semiconductor fin **815**. The structure is shown immediately following a CMP that removes excess metal from the surface **215E**. In addition to the first metal gate layer **803** and the second metal gate layer **805**, the surface **215E** exposes a high-x dielectric layer **807**, spacers **809**, and dielectric fill **811**. The high-K dielectric layer **807**, the first metal gate layer **803**, and the second metal gate layer **805** may wrap around the semiconductor fin **815** and source/drain regions **813** may be formed on the semiconductor fin **815** to provide a transistor structure. The surface **215E** may be scanned using AFM or C-AFM. C-AFM may be particularly effective for determining an extent to which metal residues are present on the surface **215E**. A remedial process may include additional CMP.

(44) In some embodiments, the first process is an etch process that recesses a dielectric adjacent a semiconductor fin. FIG. 9 illustrates a cross section **901** of a substrate **105C** immediately after a plasma etching process that causes a dielectric **905** to be recessed adjacent semiconductor fins **903**. The semiconductor fins **903** may have been grown from a bulk semiconductor **907** within trenches that were formed in the dielectric **905**. The surface **215F** may be scanned using AFM or C-AFM. AFM may be effective for determining a depth to which the dielectric **215G** has been recessed. C-AFM may provide an additional advantage by distinguishing between the dielectric **905** and the semiconductor fins **903**. The diagnostic determination may determine if the semiconductor fins **903** have been formed properly or whether the dielectric **905** has been recessed to the correct depth. In some embodiments, a remedial process includes additional etching to further recess the dielectric **905**. In some embodiments, a remedial process includes redeposition and etching of the dielectric **905**. In some embodiments, a remedial process includes polishing away and regrowing the semiconductor fins **903**.

(45) C-AFM may be particularly effective for distinguishing between dielectrics, semiconductors, and metal structures on a surface. Examples of dielectrics include, without limitation, silicon oxide (SiO₂), silicon nitride (SiN), silicon carbide (SiC), silicon carbonitride (SiCN), silicon

oxycarbide (SiOC), silicon oxycarbonitride (SiOCN), low-K dielectrics, extremely low-K dielectrics, and the like. A low-k dielectric is a material having a smaller dielectric constant than SiO₂. SiO₂ has a dielectric constant of about 3.9. Examples of low-k dielectrics include, without limitation, organosilicate glasses (OSG) such as carbon-doped silicon dioxide, fluorine-doped silicon dioxide (FSG), organic polymer low-k dielectrics, porous silicate glass, and the like. An extremely low-k dielectric is a material having a dielectric constant of about 2.1 or less. An extremely low-k dielectric may be a low-k dielectric with additional porosity. A high-k dielectric is a material having a smaller dielectric constant than SiO₂. A high-x dielectric may be a metal oxide or a silicate of hafnium (Hf), aluminum (Al), zirconium (Zr), lanthanum (La), magnesium (Mg), barium (Ba), titanium (Ti), lead (Pb), or the like. Examples of high-x dielectrics include, without limitation, titanium oxide (TiO₂), hafnium zirconium oxide (HfZrO), tantalum oxide (Ta₂O₃), hafnium silicon oxide (HfSiO₄), hafnium oxide (HfO₂), zirconium oxide (ZrO₂), zirconium silicon (ZrSiO₂), and the like. Examples of conductive metals include, without limitation, tantalum, titanium, platinum, gold, iridium, tungsten, nickel, ruthenium, copper, nitrides thereof, silicides thereof, alloys thereof, and the like.

(46) In some embodiments, scanning surveys only a portion of the substrate surface. In some embodiments, the portion of the surface that is surveyed is 10^{sup}−6 percent or less a total area of the surface. In some embodiments, the portion of the surface that is surveyed is 10^{sup}−8 percent or less a total area of the surface. In some embodiments, the portion of the surface that is surveyed is from 0.01 to 100 μm^{sup}2. In some embodiments, the portion of the surface that is surveyed is from 0.1 to 10 μm^{sup}2. The area surveyed may be one more distinct zones on the surface. In some embodiments, the area surveyed comprises two or more disjoint zones on the surface. In some embodiments, the substrate is a wafer, and a first zone is closer to a center of the wafer than a second zone. This type of zone selections may be beneficial after a process such as CMP, which may proceed at a different rate near the edge of a wafer as compared to its center. Each zone that is scanned may include a plurality of rows and columns representing discrete data points. In some embodiments, each zone includes from 10 to 10^{sup}5 rows and from 10 to 10^{sup}5 columns. In some embodiments, each zone includes from 10^{sup}2 to 10^{sup}4 rows and from 10^{sup}2 to 10^{sup}4 columns.

(47) In some embodiments, the zone scanned is varied from substrate to substrate relative to the geometry of the substrate. For example, one wafer may be scanned in a zone nearer the center and the next wafer scanned in a zone nearer the edge. In some embodiments, the area to be scanned is varied in a predetermined manner. In some embodiments, the area to be scanned is varied randomly. These strategies may increase the probability that a problem occurring in only one area of a substrate is detected before it affects a large number of substrates.

(48) In some embodiments, each zone that is scanned includes a plurality of instances of one type of device. For example, the area scanned may be a portion of a memory cell array and include a plurality of memory cells. In some embodiments, the number of devices scanned is between 2 and 10^{sup}4. In some embodiments, the number of devices scanned is between 5 and 10^{sup}3. In some embodiments, the number of devices scanned is between 10 and 500. For a fixed number of measurement points, varying a spacing between measurement points provides a tradeoff between the number of devices included in the survey and the precision with which each device is scanned. In some embodiments, the vertical and horizontal spacings between measurement points are in the range from 0.1 nm to 100 nm. In some embodiments, the vertical and horizontal spacings between measurement points are in the range from 0.2 nm to 10 nm. In some embodiments, the vertical and horizontal spacings between measurement points are in the range from 0.5 nm to 1 nm.

(49) FIG. 10 provides a plot 1001 of one row of data from a C-AFM scan of the memory array 702 of the substrate 105A illustrated by the cross-sectional view 701 of FIG. 7. FIG. 10 includes sets of peaks 1009A-1009E, each of which represents a distinct memory cell 402. A diagnostic determination as to whether a manufacturing issue is present may be made according to whether

any of the peak values is above or below a threshold value or whether a particular pattern is present in the data. For example, the group of peaks **1009A** includes a first peak **1003**, a middle peak **1007**, and a third peak **1005**. The middle peak **1007** is lower than the first peak **1003** and the third peak **1005**. This structure may indicate there is a leakage path at an edge of the memory cell **402** that corresponds to the group of peaks **1009A**.

(50) FIG. **11** provides a flow chart of a method **1100** that may be used to processing the data that is collected by scanning probe microscopy. The method **1100** begins with act **1101**, which is gathering the data. Act **1103** is forming the data into an image. FIG. **12** provides an example. FIG. **12** shows an image **1201** that represents all the data from the C-AFM scan of the memory array **702** of the substrate **105A** illustrated by the cross-sectional view **701** of FIG. **7** as a grey-scale image. In some embodiments, a plurality of images are formed due to there being a plurality of data types. Transforming the data may include operations such as filtering, averaging, scaling, and interpolating. The results may be stored as arrays of pixels, each pixel having a value representing a measurement value at a particular (X, Y) coordinate. If there are multiple data types, e.g., topography and conductance data, it is desirable that the images corresponding to these different data types each have the same number pixels with values for corresponding coordinates.

(51) The method **1100** may continue with act **1105**, identifying portions of the data images that correspond to individual devices. In the image **1201**, these portions are device images **1203**, which correspond to distinct memory cells **402**. The art of image processing provides algorithms adapted to this purpose. These algorithms may be referred to as “blob detection” or “interest point detection” methods. Once identified, the device images **1203** may be analyzed, which is act **1107**, and the analysis used to make a diagnostic determination, which is act **1109**. In some embodiments, corresponding areas of a second image representing another data type are analyzed to make the diagnostic determination. For example, a C-AFM image may be used to identify the device locations and an AFM image used to evaluate device characteristics, or vice versa. In some embodiments, the diagnostic determination is made according to just the data in the portions of the data image that relate to particular devices. A diagnostic determination may relate to such matters as whether electrodes are adequately exposed, whether leakage currents are excessive, or whether trenches are adequately deep, and the like. In some embodiments a diagnostic determination relates to whether all the examined devices have an electrical property within a predetermined specification.

(52) In some embodiments, a diagnostic determination is made based on the device shapes. For this purpose, the image may be transformed virtually or otherwise from a greyscale image to a black and white image. The transformation from the image **1201** of FIG. **12** to the image **1301** of FIG. **13A** provide an example. The device images **1303** of image **1301** are formed from the device images **1203** of FIG. **1201**. In some embodiments, forming the black and white device images from the greyscale device images comprises stripping one or more outer pixels from the greyscale device images. In some embodiments, an outermost region of one to ten pixels width is stripped from the grey scale image. For example, the device images **1303** lack the outermost two pixels of the device images **1203**. Stripping these eccentric pixels may reduce noise in the shape data. The outermost pixels may be determined based on distance from a centroid. Alternatively, the outermost pixels may be determined based on distance from a perimeter. In some embodiments, a diagnostic determination is made according to the areas of the device images **1303**.

(53) In some embodiments, a diagnostic determination is made based on the conformation of the perimeters of the device shapes. For this purpose, the device image centers may be removed. These centers may be obtained by selecting pixels that are more than a certain number of pixels from any outer edge. For example, the image **1311** of FIG. **13B** has center areas **1313** that correspond to those pixels of the device images **1303** that are five pixels or more from outside the device images **1303**. Subtracting the image **1311** from the image **1301**, or equivalently, subtracting the center areas **1313** from the device images **1303**, provides the image **1321** of FIG. **13C**, which contains

device perimeters **1323**. The device perimeters **1323** may be analyzed for a property such as circularity and a diagnostic determination may be made on that basis. Alternatively, or in addition, the center areas **1313** may be analyzed for a property such as circularity and a diagnostic determination may be made on that basis.

(54) FIG. **14** provides a flow chart of a method **1400** that may be used in the manufacturing system **100**. The method **1400** is similar to the method **300** and contains many of the same steps. Principal differences are that method **1400** uses the diagnostic determination of act **307** in act **1401**, determining if the substrate has a defect and the method **1400** has act **1403**, disposing of or recycling the substrate if a defect is detected. Early detection and disposal of substrates with defects can be useful in increasing overall throughput and reducing costs associated with complete manufacturing of substrates that do not contribute to yield.

(55) FIG. **15** provides a flow chart of a method **1500** that may be used in the manufacturing system **100**. The method **1500** is similar to the method **300** and contains many of the same steps. A principal difference is that method **1500** uses the diagnostic determination of act **307** in act **1503**, determining if there is a fault in a process used to produce the substrate and act **1501**, performing process tool maintenance or another process adjustment in response to the detections. In some embodiments a sputtering target is inspected or replaced based on the diagnostic determination. In some embodiments an edge ring or other consumable part in a plasma chamber is inspected or replaced based on the diagnostic determination. In some embodiments, a tool is cleaned based on the diagnostic determination.

(56) While the method **300** of FIG. **3**, the method **1400** of FIG. **14**, and the method **1500** of FIG. **15** have been illustrated and described herein as a series of acts or events, it will be appreciated that the illustrated ordering of such acts or events is not to be interpreted in a limiting sense. For example, some acts may occur in different orders and/or concurrently with other acts or events apart from those illustrated and/or described herein. Further, not all illustrated acts are required to implement one or more aspects or embodiments of the description herein, and one or more of the acts depicted herein may be carried out in one or more separate acts and/or phases.

(57) Some aspects of the present teachings relate to a method that includes applying a first manufacturing process to a substrate having a surface, scanning a portion of the surface using a scanning probe microscope to obtain data, processing the data to make a diagnostic determination, and selectively applying a second manufacturing process to the substrate based on the diagnostic determination.

(58) Some aspects of the present teachings relate to a method that includes receiving a substrate having a plurality of device structures, scanning a portion of the substrate using a scanning probe microscope to obtain data, forming one or more images from the data, identifying areas of the one or more images that correspond to individual devices among the plurality of device structures, applying an analysis to the areas that correspond to individual devices, and using a result of the analysis in a manufacturing process or system.

(59) Some aspects of the present teachings relate to an integrated circuit device manufacturing system that includes a first processing tool, a second processing tool, a substrate handling system, a scanning probe microscope, and a computer processor. The scanning probe microscope receives substrates from the first processing tool. The computer processor receives data from the scanning probe microscope. The substrate handling system selectively delivers substrates to the first manufacturing processing tool according to instructions from the computer processor.

(60) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that

they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method comprising: applying a first manufacturing process sequence to produce an array of memory cells over a semiconductor substrate, wherein the memory cells comprise a top electrode, a bottom electrode, and a magnetic tunnel junction between the top electrode and the bottom electrode; scanning a tip of a conductive atomic force microscope across a surface of the semiconductor substrate at which the top electrodes are at least partially exposed to obtain a first data set and a second data set, wherein the first data set corresponds to deflection of the tip and relates to variations in height of the surface and the second data set corresponds to conduction between the tip and the semiconductor substrate and relates to variations in conductivity across the surface; using the first data set and or the second data set to identify areas of the surface that contain individual memory cells; and analyzing at least the second data set in the areas that contain individual memory cells to determine whether defects are present, and selectively applying further processing to the semiconductor substrate according to results of the analysis.
2. The method of claim 1, wherein the analysis comprises determining shapes of individual device structures within the areas that contain individual memory cells.
3. The method of claim 2, wherein determining shapes of individual device structures within the areas that contain individual memory cells comprises forming a grey-scale image from the first or second data set and analyzing the grey-scale image using an image processing algorithm.
4. The method of claim 1, wherein the analysis is based on shapes of perimeters of the areas that correspond to individual device structures.
5. A method for analyzing a semiconductor substrate during an intermediate stage of manufacture, the method comprising: processing the semiconductor substrate using a first manufacturing process that forms devices with electrodes exposed at a surface of the semiconductor substrate; scanning the surface using a conductive atomic force microscope (C-AFM) to obtain both topography data and conductivity data, wherein the topography data relates to deflection of a tip of the C-AFM by the surface and the conductivity data relates to current between the tip and the semiconductor substrate; using the topography data or conductivity data to identify areas containing individual electrodes; determining shapes of exposed portions of the individual electrodes based on the conductivity data; making a diagnostic determination based on the determined shapes of the exposed portions of the individual electrodes; and selectively applying or omitting a second manufacturing process based on the diagnostic determination.
6. The method of claim 5, wherein the second manufacturing process is a remedial process selectively performed based on the diagnostic determination.
7. The method of claim 5, wherein the first manufacturing process comprises a first chemical mechanical polishing (CMP) that exposes the electrodes.
8. The method of claim 7, wherein the second manufacturing process comprises additional CMP and is selectively applied when the diagnostic determination is that the first CMP did not adequately expose the electrodes.
9. The method of claim 5, wherein the first manufacturing process comprises a spacer etch.
10. The method of claim 9, wherein the second manufacturing process comprises a spacer re-etch and is selectively applied when the diagnostic determination is that the spacer etch did not adequately expose the electrodes.
11. The method of claim 5, wherein the first manufacturing process comprises ion beam etching that shapes the electrodes.
12. The method of claim 5, wherein the second manufacturing process is a wet clean and is selectively applied when the diagnostic determination is that there is excessive conductive residue

on the surface.

13. The method of claim 6, wherein the remedial process comprises an additional etching, deposition, or cleaning process.

14. The method of claim 5, further comprising performing a third manufacturing process in parallel with making the diagnostic determination.

15. The method of claim 5, wherein the electrodes are top electrodes of MRAM cells.

16. The method of claim 5, wherein the electrodes are top electrodes of fin field effect transistors.

17. The method of claim 5, further comprising controlling a substrate handling system to selectively route the semiconductor substrate for further processing, remedial processing, or disposal based on the diagnostic determination.

18. The method of claim 5, further comprising selectively discarding substrates based the diagnostic determination.

19. A method for analyzing a semiconductor substrate during an intermediate stage of manufacture, the method comprising: processing the semiconductor substrate using a first manufacturing process that forms devices with electrodes exposed at a surface of the semiconductor substrate; scanning the surface using a conductive atomic force microscope (C-AFM) to obtain both topography data and conductivity data, wherein the topography data relates to deflection of a tip of the C-AFM by the surface and the conductivity data relates to current between the tip and the semiconductor substrate; using the topography data or the conductivity data to identify areas containing individual devices; determining an electrical property for each of the individual devices from the conductivity data; making a diagnostic determination based on the electrical property; and selectively applying or omitting a second manufacturing process based on the diagnostic determination.

20. The method of claim 19, wherein the areas containing individual devices are identified from the topography data.
